

The LNM Institute of Information Technology, Jaipur

Department of Electronics and Communication Engineering

Microwave Lab Experiment report – 2025-2026

Experiment No.: 1

REFLEX KLYSTRON CHARACTERISTICS

Name: - Priyanshi Jain

Roll No.: - 24UEC249

Day: Friday

Date of Experiment: 30 Jan 2026

Faculty / Lab Instructor Name: Mr. Kamta Sharma

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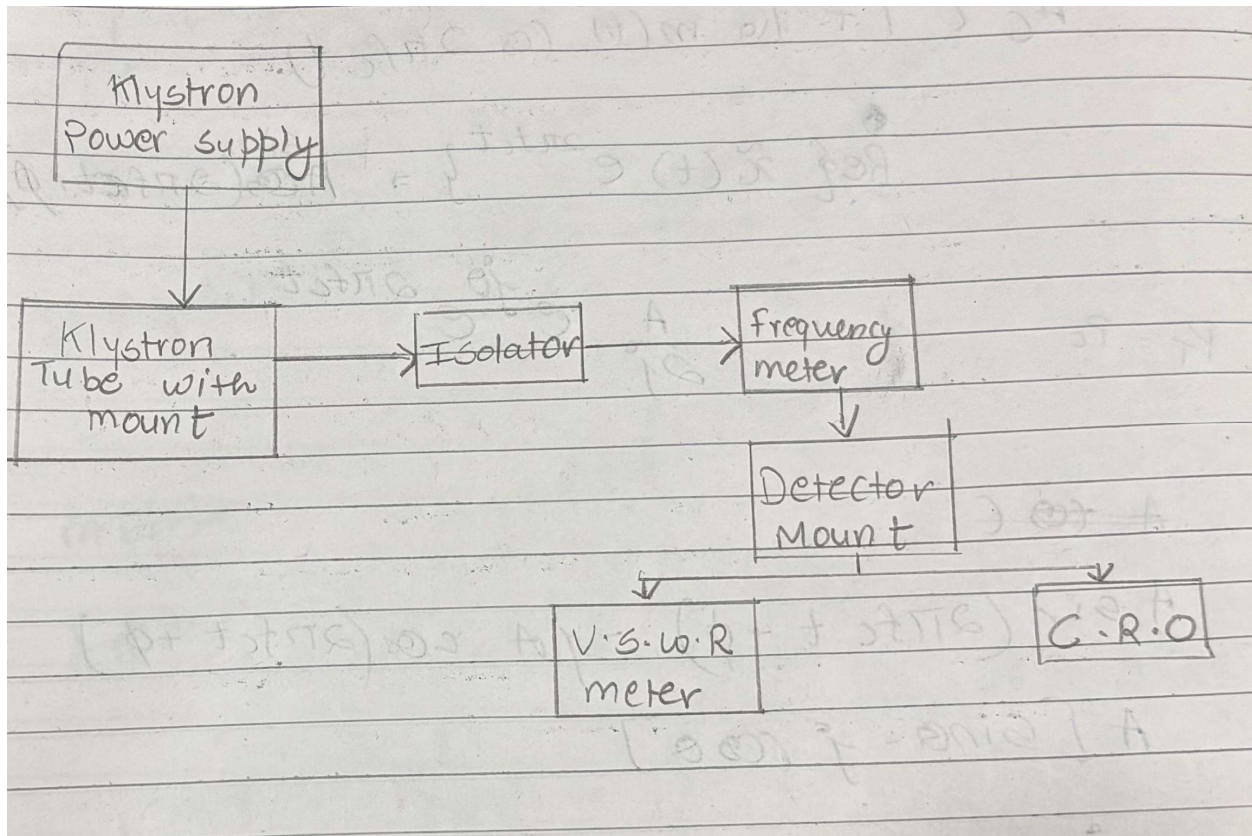
1) Aim:

To study mode characteristics of reflex klystron and hence to determine mode number, transit time electronic tuning range (ETR), and electronic tuning sensitivity (ETS)

2) Equipment and Component:

1. Klystron power supply
2. Klystron tube /2k25
3. Isolator
4. Frequency Meter
5. Variable Attenuator
6. Detector mount
7. Waveguide stands
8. VSWR meter
9. Cathode Ray Oscilloscope (CRO)
10. Tool Kit (Screwdriver, Wrench, Screws & Nuts etc.)

3)Block Diagram



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4) Theory

- Reflex klystron is a microwave vacuum tube used as a microwave source, operating on the principle of velocity modulation.
- Electrons emitted from the cathode are accelerated toward the resonator grids and then repelled by a negatively biased reflector, causing them to reverse direction and pass through the grids again.
- An RF electric field exists across the resonator grids during oscillation, which accelerates or retards electrons depending on the instantaneous grid voltage.
- Due to different velocities, electrons take different transit times in the reflector space and form electron bunches—this process is called velocity modulation.
- When the returning electron bunches pass through the grids during a retarding RF phase, they transfer energy to the resonator, sustaining oscillations.
- Maximum oscillation occurs when the electron transit time equals $(n+3/4)$ cycles of the resonant frequency, where n is an integer; otherwise, oscillations are not sustained.

5) Procedure

1. Connect the components and equipment as shown in block diagram.
2. Keep the control knob of the Klystron power supply as shown below:
 Mode switch: AM
 Beam voltage knob: Fully anticlockwise
 Repeller voltage knob: Fully clockwise
 Meter switch: Beam Voltage/Beam Current/ Repeller Voltage: Beam current
3. Rotate the frequency meter at one side (NOTE: Rotate the frequency meter very slowly).
4. Switch on the Klystron power supply, V.S.W.R/CRO, and cooling fan for the Klystron tube.
 Wait for 1-2 minutes for the klystron to respond.
5. Beam Voltage (Cathode voltage) knob at minimum position gives a beam voltage of about 250V (Max Range 300V). Observe the beam current on the meter by changing the meter switch to the beam current position. “The beam current should not be more than 30mA”. (Try to set the Beam current at less than 30 mA by increasing/Decreasing the beam voltage knob.)
6. Use AM amplitude and frequency controls in the Klystron power supply to observe the clear square waveform on the Oscilloscope (CRO) front panel.
7. Now change the meter switch (Selector switch) to the repeller/reflector voltage position.
8. Decreasing the reflector/Repeller voltage, record the output voltage
9. To measure frequency: Rotate the frequency meter, observe a dip in the output voltage on the CRO, and note the corresponding frequency meter reading between two horizontal red lines.
10. Change the repeller voltage and repeat steps 8 and 9 for each repeller voltage.

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7) Faculty-Evaluated Observation Sheet

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Microwave Lab Experiment observation report – 2025-2026

Exp. Name: Reflex Klystron

Name: PRIYANSHI JAIN Date: 30-01-26

Roll No. 24NEC249 Day: Friday

Marks & Invigilator Signature

Mode	S.No.	Freq. (GHz)	Repeller voltage (V)	Output (V ²)
①	1. At Center	8.865	<u>324</u>	2.2
	2. Half Right	8.88	347	1.1
	3. Half Left	8.845	312	1.1
	4. Mid Right	8.87	345	1.5
	5. Mid Left	8.855	316	1.5
	6. Zero Right	0	350	0
	7. Zero Left	0	309	0
②	1. At Center	8.865	<u>224</u>	1.6
	2. Half Right	8.895	234	0.8
	3. Half Left	8.84	202	0.8
	4. Mid Right	8.885	232	1
	5. Mid Left	8.85	208	1
	6. Zero Right	0	239	0
	7. Zero Left	0	199	0

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Mode	S.No.	Freq. (GHz)	Repeller Voltage (Vr)	O/P (V _o) ²
(3)	1. At Center	8.84	92	0.7
	2. Half Right	8.855	95	0.35
	3. Half Left	8.8763	84	0.35
	4. Mid Right	8.845	93	0.5
	5. Mid Left	8.879	86	0.5
	6. Zero Right	0	98	0
	7. Zero Left	0	82	0

(*) CALCULATION

Modes	ETR (GHz)	ETS (MHz/V)
1.	8.84 to 8.88 GHz	$\frac{8.88 - 8.845}{345 - 312}$ $= 0.001 \text{ GHz/V}$ $= 1 \text{ MHz/V}$
2.	8.84 to 8.895 GHz	$\frac{8.895 - 8.84}{237 - 202}$ $= 1.718 \text{ MHz/V}$
3.	8.763 to 8.855 GHz	$\frac{8.855 - 8.763}{95 - 84}$ $= 8.36 \text{ MHz/V}$

(10)

30/11/26

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8) Calculation

② CALCULATION

Modes	ETR (GHz)	ETS (MHz/V)
1.	8.84 to 8.88 GHz	$\frac{8.88 - 8.845}{345 - 312}$ $= 0.001 \text{ GHz/V}$ $= 1 \text{ MHz/V}$
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3.	8.763 to 8.855 GHz	$\frac{8.855 - 8.763}{95 - 84}$ $= 8.36 \text{ MHz/V}$

⑩ 30/10/26

9) Result

Modes	ETR(GHz)	ETS(MHz/V)
Mode 1	8.84-8.88	1
Mode 2	8.84-8.895	1.718
Mode 3	8.763-8.855	8.3

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10) Conclusion

By optimizing the beam and repeller voltages, the reflex klystron was successfully operated as a microwave source. To ensure stable oscillations, the beam current was maintained at 14mA, well below the 30mA threshold. It was observed that the output power and oscillation frequency were highly sensitive to fluctuations in the repeller voltage. Frequency measurements were conducted using a cavity frequency meter, identified by characteristic dips in the CRO trace. These results successfully demonstrated the electronic tuning characteristics of the reflex klystron, with measured values for Electronic Tuning Range (ETR) and Electronic Tuning Sensitivity (ETS) aligning closely with theoretical predictions.